

hey today lets focus on week 11 lesson - In this lesson, you will be able to learn Bayesian Nets. On successful completion of this lesson, you will be able to,

Describe basic techniques used for Bayesian Learning
Compute the prior and posterior of a distribution
Predictive distribution of an event using Bayesian Learning

Lesson Outline:

Introduction to Bayesian Learning
Prior and Posterior Distributions
Bayesian learning for Predictive distribution

That's a *really* good topic to focus on — Bayesian Nets are one of those things that feel confusing at first, but once it clicks, it becomes super intuitive (and honestly kind of satisfying 😊). Let's go step by step in a **simple, exam-friendly way** so you feel confident.

1. Introduction to Bayesian Learning

Core Idea:

Bayesian Learning is all about **updating your belief when you get new data**.

Think of it like this:

- You have an **initial guess** (prior belief)
- You observe **new evidence (data)**
- You update your belief → get a **better estimate**

 This is based on Bayes' Theorem

Real-life intuition:

Imagine:

- You think it **might rain today (50%)**
- Then you see **dark clouds**
- Now your belief increases → maybe **80% chance of rain**

That update = Bayesian Learning.



2. Prior and Posterior Distributions

This is the heart of the lesson.

● Prior (Before seeing data)

- Your **initial belief**
- Written as:
 $P(H)$ → Probability of hypothesis H

👉 Example:

- Probability a student passes = 0.6
(before seeing study hours)
-

● Posterior (After seeing data)

- Updated belief after observing data
 - Written as:
 $P(H | D)$ → Probability of H given data D
-



The Formula (VERY IMPORTANT)

$$P(H | D) = \frac{P(D|H) \cdot P(H)}{P(D)}$$



What each part means:

- $P(H)$ → Prior
- $P(D | H)$ → Likelihood (how likely data is if H is true)
- $P(D)$ → Evidence (normalizing constant)
- $P(H | D)$ → Posterior

Simple Example:

Suppose:

- 70% students study $\rightarrow P(H) = 0.7$
- If a student studies, 90% pass $\rightarrow P(D | H) = 0.9$

Then we update belief about passing based on study behavior.

3. Predictive Distribution (Final Goal)

This is where Bayesian learning becomes powerful.

Idea:

We don't just predict using one fixed model —
we consider **all possible hypotheses** and weigh them.

Formula:

$$P(x_{new} | D) = \sum_h P(x_{new} | h) \cdot P(h | D)$$

Meaning:

To predict a new event:

- Consider all hypotheses h
 - Multiply:
 - Probability of prediction under h
 - Posterior probability of h
 - Then sum them
-

Intuition:

Instead of saying:

"This is the answer"

Bayesian says:

"These are all possible answers, weighted by how likely they are"

4. How Bayesian Nets Fit In

A **Bayesian Network** is just:

- A **graph** (nodes + arrows)
- Showing **probabilistic relationships**

👉 Example:

- Study → Exam Score → Pass/Fail

Each node = variable

Each arrow = dependency

Why use them?

- Handle uncertainty
 - Work well with incomplete data
 - Easy to visualize dependencies
-

Exam Tips (VERY IMPORTANT)

If they ask:

✓ "Define Bayesian Learning"

Say:

A method of updating prior beliefs using observed data to obtain posterior probabilities.

✓ "Prior vs Posterior"

- Prior = before data
- Posterior = after data

✓ “Predictive distribution”

Weighted sum of predictions over all hypotheses using posterior probabilities.

💬 If you want next step:

We can:

- Do past paper questions
- Solve numerical problems step-by-step
- Or I can explain **Bayesian Networks with a diagram + example question**

Just tell me — we’ll make this super clear until you feel 100% confident 🎯



how is this different from naive bayes

Ahh this is a *very smart question* — this is exactly where many students get confused. Once you see the difference, everything becomes much clearer 😊

🧠 Big Picture First

👉 Bayesian Learning = a *general framework*

👉 Naive Bayes = a *specific algorithm/model* built using that framework

Think of it like:

- Bayesian Learning = **the theory**
 - Naive Bayes = **one simple application of that theory**
-

🔍 1. Bayesian Learning (General Idea)

💡 What it does:

- Updates probabilities using **Bayes' Theorem**
 - Works with:
 - Priors
 - Posteriors
 - Predictive distributions
-

Key Feature:

- ✓ Considers many possible hypotheses
- ✓ Combines them using probabilities

👉 Very flexible, but:

- Can be computationally expensive
 - More theoretical/general
-

2. Naive Bayes (Specific Algorithm)

What it does:

- Uses Bayes' Theorem for **classification**
- Makes a BIG assumption:

All features are **independent** of each other

Formula used:

$$P(C \mid x_1, x_2, \dots, x_n) \propto P(C) \prod_{i=1}^n P(x_i \mid C)$$

Key Feature:

- ✓ Assumes:
 - Feature 1 $\perp$ Feature 2 $\perp$ Feature 3 (independent)

👉 Example:

- Words in an email are treated independently in spam detection

 3. Side-by-Side Difference

Aspect	Bayesian Learning	Naive Bayes
Type	General framework	Specific algorithm
Goal	Update beliefs, prediction	Classification
Hypotheses	Many hypotheses	One model
Feature dependency	Can model dependencies	Assumes independence
Complexity	High	Low (fast)
Real-world use	Theoretical + advanced models	Very common (spam, text classification)

 Simple Intuition

 Bayesian Learning:

“Let’s consider *all possibilities* and weigh them properly”

 Naive Bayes:

“Let’s simplify things and assume features don’t affect each other”

 Example to Lock It In

 Problem: Predict if student passes

Bayesian Learning:

- Considers:
 - Study hours
 - Attendance
 - Sleep

- Also considers **relationships between them**
-

Naive Bayes:

- Assumes:
 - $\text{Study} \perp \text{Attendance} \perp \text{Sleep}$
 - Treats them **separately**
-



Why “Naive”?

Because the independence assumption is often **not true in real life**

BUT...



It still works surprisingly well!



Exam-ready answer (short)

If they ask:



Difference between Bayesian Learning and Naive Bayes:

Bayesian Learning is a general probabilistic framework that updates beliefs using Bayes' theorem, while Naive Bayes is a specific classification algorithm that applies Bayes' theorem under the assumption that features are conditionally independent.

If you want next, we can:

- Do a **numerical Naive Bayes example (step-by-step)** 
- Or I can show how **Bayesian Networks remove the “naive” assumption** (this is  for exams)

Just tell me 



is bayes net and bayesian learning same

Ooo this is another really important distinction — and you're *very close* to mastering this whole topic now 😊

Short answer:

- ✗ No, they are NOT the same
- ✓ But they are closely related

Let's break it down clearly so it sticks.

Big Idea

👉 **Bayesian Learning** = *the method (how we learn/update probabilities)*

👉 **Bayesian Network** = *the model (how we represent variables and relationships)*

1. Bayesian Learning (The Process)

What it is:

A mathematical approach that uses Bayes' Theorem to update beliefs.

What it does:

- Starts with **Prior**
 - Uses **Data**
 - Produces **Posterior**
 - Can compute **predictive probabilities**
-

Think of it like:

"How do we *learn from data* and update our beliefs?"



2. Bayesian Network (The Model)



What it is:

A **graphical model** that represents:

- Variables (nodes)
 - Dependencies (arrows)
-



Example structure:

- Study → Exam Score → Pass
 - Sleep → Exam Score
-



What it does:

- Shows **who depends on whom**
 - Helps compute probabilities efficiently
-



Think of it like:

"How are variables *connected* and influence each other?"



3. How They Connect

This is the key part 📌



You can use **Bayesian Learning** inside a Bayesian Network

Meaning:

- Network gives the **structure**
- Learning updates the **probabilities in that structure**



4. Side-by-Side Difference

Aspect	Bayesian Learning	Bayesian Network
Type	Method / approach	Model / structure
Purpose	Update probabilities	Represent relationships
Based on	Bayes' theorem	Graph theory + probability
Output	Posterior probabilities	Graph with probabilities
Usage	Learning from data	Modeling dependencies



Simple Analogy

Imagine:



Bayesian Network:

A **map** of a city 🗺️ (shows roads and connections)



Bayesian Learning:

A **GPS system** 🚗 (updates best route based on traffic/data)



In Your Assignment Context

For your ML assignment:

- If you're **calculating prior/posterior** → **Bayesian Learning**
- If you're **drawing nodes and arrows** → **Bayesian Network**



Exam-ready answer

Bayesian Learning is a probabilistic method used to update beliefs using Bayes' theorem, whereas a Bayesian Network is a graphical model that represents probabilistic

relationships among variables. Bayesian learning can be applied within a Bayesian network to update its probabilities.

If you want, next we can:

- Build a **small Bayesian Network example step-by-step** (very likely exam question 🔥)
- Or solve a **posterior probability problem together**

Just say the word 👍



hey i didnt' get this part 🤔 3. Predictive Distribution (Final Goal)

This is where Bayesian learning becomes powerful.

💡 Idea:

We don't just predict using one fixed model — we consider all possible hypotheses and weigh them. do the model change the H (the prior) or the D all possible new data, as to sentence it seems like H but why and how H supposed to change , wont it be fixed

I *love* this question — this is exactly the point where most people get stuck, and once it clicks, Bayesian learning suddenly feels very natural 😊

Let's slow it down and clear the confusion properly.

💡 First — what is actually changing?

You asked:

Does the model change H (hypothesis) or D (data)?

👉 Answer:

- **Data (D)** = fixed (what you observed)
 - **Hypotheses (H)** = a *set of possible explanations*
 - What changes is 👉 **our belief about each hypothesis**
-

Key Correction to Your Thinking

You said:

“Won’t H be fixed?”

👉 This is the *main misunderstanding*

✗ In Bayesian learning, **H is NOT just one fixed thing**

✓ Instead, we consider **many possible hypotheses**

What is H really?

H = a possible “model” or “explanation”

📌 **Example:**

Let’s say we want to predict if a student passes.

Possible hypotheses (H):

- H_1 : “Study hours matter a lot”
- H_2 : “Attendance matters more”
- H_3 : “Both matter equally”

👉 These are **different models**

Before seeing data

We assign priors:

- $P(H_1) = 0.3$
 - $P(H_2) = 0.4$
 - $P(H_3) = 0.3$
-



After seeing data (D)

Using Bayes' Theorem, we update:

- $P(H_1 \mid D)$
- $P(H_2 \mid D)$
- $P(H_3 \mid D)$



So:

- ✓ H itself doesn't change
- ✓ The probability of each H changes



Now — Predictive Distribution (Your Confusion Part)

Here's the formula again:

$$P(x_{new} \mid D) = \sum_h P(x_{new} \mid h) \cdot P(h \mid D)$$



What this REALLY means

Instead of saying:

"This one hypothesis is correct"

We say:

"Each hypothesis has some probability — let's combine them"



Step-by-step intuition

Step 1:

Each hypothesis predicts something:

- $H_1 \rightarrow$ predicts pass = 0.8
- $H_2 \rightarrow$ predicts pass = 0.6
- $H_3 \rightarrow$ predicts pass = 0.7

Step 2:

Weight them using posterior:

- $P(H_1 | D) = 0.2$
- $P(H_2 | D) = 0.5$
- $P(H_3 | D) = 0.3$

Step 3:

Combine:

👉 Final prediction =

- $(0.8 \times 0.2) + (0.6 \times 0.5) + (0.7 \times 0.3)$



So what changed?

Thing	Changes?
Data (D)	✗ No
Hypotheses (H)	✗ No
Belief about H	✓ YES (this is key!)



Why do we do this?

Because we are **uncertain** which hypothesis is correct.

Instead of:

picking ONE model

We:

average over ALL models (smart decision)

🔥 Final Insight (Very Important)

👉 Naive Bayes:

- Picks ONE model
- Uses it directly

👉 Bayesian Learning:

- Keeps MANY models
- Combines them using probabilities

💬 One-line clarity

Hypotheses (H) don't change — only their probabilities change, and predictive distribution combines all hypotheses weighted by their posterior probabilities.

If you want, next I can:

- ↳ Turn this into a numerical exam question and solve it with you
- Or ↳ connect this directly to Naive Bayes vs full Bayesian prediction

You're *very close* to mastering this — just one more push 🏆



here what is x_{new} in the formula can show fit a example to the formula

Yesss this is the *perfect* follow-up — once you understand **what x_{new} is**, the whole predictive distribution becomes crystal clear 😊

🧠 First — what is x_{new} ?

👉 x_{new} = a new data point we want to predict

- It's **NOT** part of the training data (D)

- It's something **unseen**
-

Example:

You trained your model using past students (that's **D**)

Now a **new student** comes:

- Study hours = 5
- Attendance = 80%

 This new student = x_{new}

The Formula (with meaning)

$$P(x_{new} \mid D) = \sum_h P(x_{new} \mid h) \cdot P(h \mid D)$$

What are we computing?

 "Given what we learned (**D**), what is the probability of this new case?"

Full Example (Step-by-Step)

Let's make it very concrete.

Problem:

Predict if a new student passes

So:

 x_{new} = "new student passes"

Step 1: Define hypotheses (models)

Let's say:

- H_1 : Study is most important
 - H_2 : Attendance is most important
-



Step 2: Posterior probabilities (after seeing data D)

- $P(H_1 | D) = 0.6$
 - $P(H_2 | D) = 0.4$
-



Step 3: Each hypothesis predicts for the new student

- $P(\text{pass} | H_1) = 0.8$
 - $P(\text{pass} | H_2) = 0.5$
-



Step 4: Plug into formula



$$P(x_{new}|D) = (0.8 \times 0.6) + (0.5 \times 0.4)$$



Final Answer:

$$= 0.48 + 0.20 = 0.68$$



Final Interpretation



The probability that the new student passes = 0.68 (68%)



What just happened?

Instead of:

"Use one model"

We did:

"Use ALL models, weighted by how good they are"



Quick Summary

Symbol	Meaning
D	Old data (training data)
H	Possible models/hypotheses
x_{new}	New case to predict
$P(H$	D)
$P(x_{new}$	H)



One-line clarity

x_{new} is the new unseen data point (or outcome) we want to predict using all hypotheses weighted by their posterior probabilities.

If you want next, we can:

- ↳ Do a Naive Bayes version of this same example (you'll see the difference instantly 🔥)
- Or ↳ I can give you a past paper-style question to try yourself

You're really getting into the deep understanding now — this is exactly how toppers think

100



hey now lets unpack what is in my lecture materials. - in the first two slides we have something like expectation ,

variance , and indicator variable could you explain each term decoding its formula

Yesss, this is a great place to pause and *decode the basics* — once expectation, variance, and indicator variables click, everything in Bayesian learning becomes much easier 😁
Let's unpack each one **slowly + intuitively + with formulas**.

1. Expectation (Mean / Average)

 **Idea:**

 Expectation = **the average value you expect in the long run**

Formula (Discrete case)

$$E[X] = \sum_x x \cdot P(X = x)$$

 **How to read it:**

- Take each possible value x
- Multiply by its probability
- Add everything

Example:

Let's say:

X (value)	Probability
1	0.5
2	0.3
3	0.2

👉 Compute:

$$E[X] = (1 \times 0.5) + (2 \times 0.3) + (3 \times 0.2) \\ = 0.5 + 0.6 + 0.6 = 1.7$$

🎯 Meaning:

👉 On average, X is 1.7

🎯 2. Variance (Spread / Uncertainty)

💡 Idea:

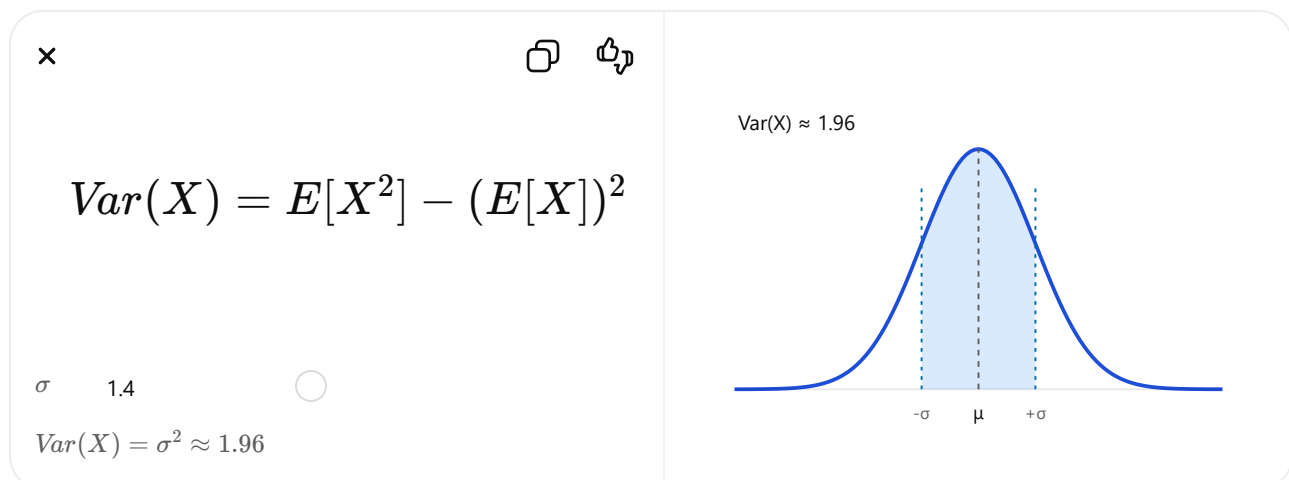
👉 Variance = how much values spread around the mean

- Low variance → values are close to mean
- High variance → values are spread out

✂ Formula:

$$\text{Var}(X) = E[(X - E[X])^2]$$

🧠 Easier version (VERY useful in exams):





Example (same data):

We already found:

- $E[X] = 1.7$
-

Step 1: Compute $E[X^2]$

$$\begin{aligned} E[X^2] &= (1^2 \times 0.5) + (2^2 \times 0.3) + (3^2 \times 0.2) \\ &= (1 \times 0.5) + (4 \times 0.3) + (9 \times 0.2) \\ &= 0.5 + 1.2 + 1.8 = 3.5 \end{aligned}$$

Step 2: Compute variance

$$\begin{aligned} \text{Var}(X) &= 3.5 - (1.7)^2 \\ &= 3.5 - 2.89 = 0.61 \end{aligned}$$



Meaning:



The spread around the mean is **0.61**



3. Indicator Variable (Very Important 🔥)

This one is *super important* in ML and probability.



Idea:



Indicator variable = a variable that tells:

- YES $\rightarrow 1$
 - NO $\rightarrow 0$
-



Notation:

$$I(A) = \begin{cases} 1 & \text{if event A happens} \\ 0 & \text{otherwise} \end{cases}$$

Example:

Let:

- $A = \text{"student passes"}$

Then:

- If pass $\rightarrow I(A) = 1$
 - If fail $\rightarrow I(A) = 0$
-

Why is this useful?

Because:

- 👉 Expectation of indicator = probability

$$E[I(A)] = P(A)$$

Example:

If:

- Probability of passing = 0.7

Then:

$$E[I(pass)] = 0.7$$

Why these matter in Bayesian Learning

Concept	Role
Expectation	Average prediction
Variance	Uncertainty in prediction

Concept

Role

Indicator

Convert events → numbers (0/1)



Super Simple Summary

- **Expectation** → average value
- **Variance** → how spread out values are
- **Indicator** → 0 or 1 for events



If you want next:

We can:

- ↳ Connect these **directly to Bayesian formulas**
- Or I can ↳ show how **indicator variables** are used in **Naive Bayes / classification**

You're building a really strong foundation here — this is exactly how ML starts to make sense ¹⁰⁰



hey i can't understand this , how does this make sense -
 $E[I(A)] = P(A)$

That confusion is *completely normal* — this is one of those things that looks strange at first, but once you see it clearly, it feels almost obvious 😊

Let's break it down very gently.



Step 1: What is $I(A)$ again?

Indicator variable:

$$I(A) = \begin{cases} 1 & \text{if event A happens} \\ 0 & \text{if event A does NOT happen} \end{cases}$$

👉 So it's just a **random variable** that can be:

- 1 (with some probability)
- 0 (with some probability)

Step 2: Think of it like a random variable X

Let's rewrite:

Let $X = I(A)$

Then:

Value of X	When	Probability
1	A happens	$P(A)$
0	A doesn't happen	$1 - P(A)$

Step 3: Now apply expectation formula

We use:

$$E[X] = \sum x \cdot P(X = x)$$

Plug values:

$$E[X] = (1 \times P(A)) + (0 \times (1 - P(A)))$$

Simplify:

$$= P(A) + 0 = P(A)$$

🌟 That's it!



$$E[I(A)] = P(A)$$



Real-life Example (Super Intuitive)

Imagine:

- Event A = "student passes"
 - Probability = 0.7
-

Indicator:

- Pass $\rightarrow 1$
 - Fail $\rightarrow 0$
-

Now think:

If you repeat this many times:

- 70% of the time $\rightarrow 1$
 - 30% of the time $\rightarrow 0$
-

Average (expectation):

$$= (1 \times 0.7) + (0 \times 0.3) = 0.7$$



Intuition (This is the key insight)



Indicator variable is like asking:

"On average, how often does this event happen?"

And that is exactly:

the probability of the event

One-line clarity

The expectation of an indicator variable equals the probability because the variable is 1 when the event occurs and 0 otherwise, so its average value equals how often the event happens.

If you want, I can next:

- ↳ Show how this is used in **machine learning loss functions**
- Or ↳ give a **quick exam-style trick question using indicator variables**

You're asking exactly the right questions — this is how deep understanding builds **100**



could you explain this properly with answer Bernoulli: a model for tossing coins - Expectation and Variance
Bernoulli r.v. x takes values in $\{0, 1\}$

1

What is the expectation of a Bernoulli Distribution?

What is the variance of a Bernoulli Distribution?

What is the distribution of N independent coin tosses?

This is a *classic* and very important piece — once you get Bernoulli properly, a lot of ML probability becomes much easier 😊

Let's go step by step and make it feel **simple + logical + exam-ready**.

1. Bernoulli Random Variable (Coin Toss Model)

 Idea:

A **Bernoulli random variable** models a situation with only **two outcomes**:

- Success $\rightarrow 1$
- Failure $\rightarrow 0$

 Example:

Coin toss:

- Heads $\rightarrow 1$
- Tails $\rightarrow 0$

Let:

- $p = P(X = 1)$ (probability of success)

Then:

- $P(X = 1) = p$
 - $P(X = 0) = 1 - p$
-

 2. Expectation of Bernoulli **Formula:**

$$E[X] = p$$

 Why is this true?

Use expectation formula:

$$E[X] = (1 \times p) + (0 \times (1 - p)) = p$$

 Meaning:

 The average outcome = probability of success

 Example:

If:

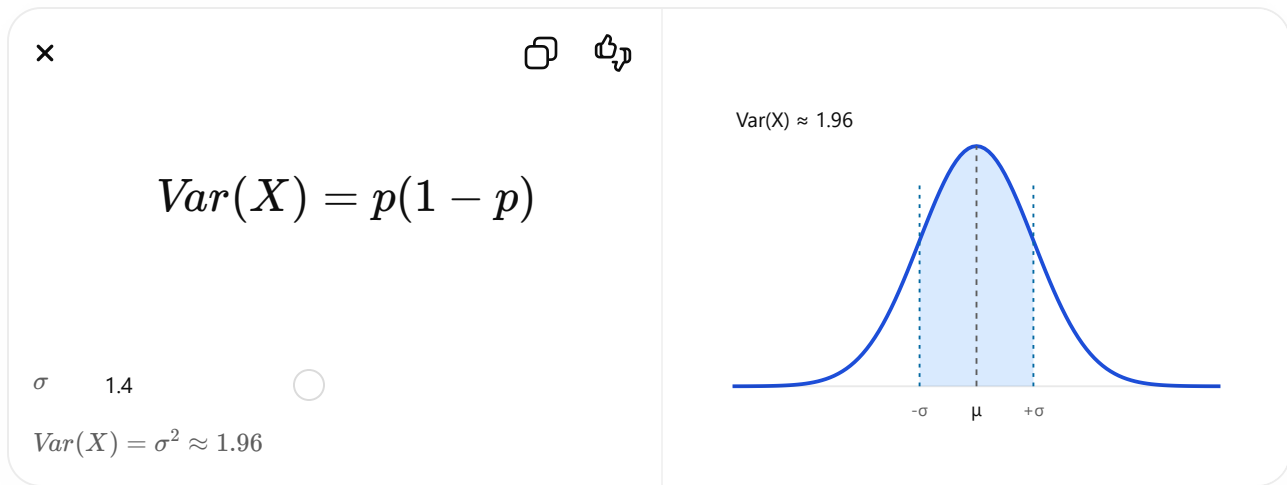
- $p = 0.7$

Then:

- On average $\rightarrow$ result = **0.7**

🎯 3. Variance of Bernoulli

📌 Formula:



🧠 Why?

Using:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

For Bernoulli:

- $X^2 = X$ (since $0^2=0$, $1^2=1$)

So:

- $E[X^2] = E[X] = p$

Then:

$$\text{Var}(X) = p - p^2 = p(1 - p)$$

🎯 Meaning:

- Measures uncertainty
- Maximum when $p = 0.5$ (most random case)

🎯 4. N Independent Coin Tosses

Now this is the next big step 🔥

💡 Idea:

You toss a coin **N times**, independently.

Each toss:

- Bernoulli(p)
-

📌 Question:

"What is the distribution of total number of successes?"

👉 Answer:

✅ Binomial Distribution

Binomial Distribution

📌 Formula:

$$P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k}$$

🧠 Meaning:

- $N \rightarrow$ number of trials
 - $k \rightarrow$ number of successes
 - $p \rightarrow$ probability of success
-

🎲 Example:

Flip coin 3 times ($p = 0.5$)

Probability of exactly 2 heads:

$$P(X = 2) = \binom{3}{2} (0.5)^2 (0.5)^1$$

$$= 3 \times 0.25 \times 0.5 = 0.375$$

Big Connection (VERY IMPORTANT)

Concept	Meaning
Bernoulli	One trial
Binomial	Many independent Bernoulli trials

Final Summary (Exam Ready)

✓ Expectation:

$$E[X] = p$$

✓ Variance:

$$Var(X) = p(1 - p)$$

✓ N independent tosses:

Follow a **Binomial distribution**

Intuition in One Line

Bernoulli models one yes/no event, and repeating it N times leads to a Binomial distribution.

If you want next, we can:

- ↳ Derive **Binomial expectation & variance** (very likely exam 🔥)
- Or ↳ connect this directly to **Naive Bayes probabilities**

You're doing *exactly* the right kind of deep learning here 💯



hey how is connecting to of us learning bayes learning

That's a *really* important question — like, this is where everything starts to connect together 😊

Right now it may feel like:

"Why are we suddenly talking about coins when we're learning Bayesian stuff??"

Let's tie it **directly** to Bayesian Learning so it finally makes sense.

Big Connection (Core Idea)

👉 Bayesian Learning = learning probabilities from data

To do that, we need:

- A model of data
- A way to update beliefs

Where Bernoulli Comes In

💡 Bernoulli = simplest probability model

It models:

- YES / NO
- 1 / 0
- Success / Failure

📌 In Bayesian terms:

We assume:

"My data comes from a Bernoulli distribution with some unknown probability p "

Now the Bayesian Part Starts

What is unknown?

 The parameter p (probability of success)

Example:

- We don't know if coin is fair
- So we don't know p

Bayesian Learning View

Instead of saying:

" $p = 0.5$ "

We say:

" p could be many possible values"

 So here:

- Hypothesis H = a possible value of p
 - $H_1: p = 0.3$
 - $H_2: p = 0.5$
 - $H_3: p = 0.8$

Now Everything Connects

Concept	In This Example
Data (D)	Results of coin tosses
Hypothesis (H)	Possible values of p
Prior $P(H)$	Belief about p before data

Concept	In This Example
Likelihood $P(D$	$H)$
Posterior $P(H$	$D)$



Example (This is the key link 🔥)

You toss a coin 5 times:

Result:

👉 H, H, T, H, T → 3 heads

Step 1: Likelihood (using Binomial)

$$P(D|p) = \binom{5}{3} p^3 (1-p)^2$$

👉 This comes directly from Bernoulli/Binomial

Step 2: Update belief

Use Bayes' Theorem

👉 Get:

- $P(p | D)$



Predictive Distribution (Final Goal)

Now we want:

👉 "What is the probability next toss = Head?"

Instead of picking one p :

We average over all possible p

🌟 So WHY Bernoulli Matters

Because it gives us:

✓ The likelihood function:

- $P(D | H)$

👉 Without this, Bayesian learning cannot work

🎯 Simple Intuition

🟡 Bernoulli:

"How does data behave?"

🟢 Bayesian Learning:

"What do we believe about the parameter after seeing data?"

🔗 One-Line Connection

Bernoulli/Binomial models define how data is generated, and Bayesian learning uses them to update beliefs about the unknown parameter p .

🧠 Even Simpler

- Bernoulli = data model
 - Bayesian Learning = learning the parameter of that model
-

If you want next, we can:

- ↳ Do a full Bayesian coin example (prior → posterior → prediction) step-by-step
- Or ↳ 🔥 connect this to Naive Bayes classification (same idea but with features)

You're asking *exactly* the questions that build real understanding — this is how it all clicks

100



can you explain this part clearly , and i can't understand the symbols in the formula here - Maximum Likelihood Estimation — Procedure
 x } write down the
 Step 1: Given n data, $atl:n$
 expression for the joint distribution of the data:
 $P(\text{Till})$
 Step 2: Compute the log-likelihood
 Step 3: Differentiate and equate to zero to find the estimate of
 What is the distribution of N independent coin tosses?
 Compute
 the MLE for Bernoulli 0
 (Hint: Take m as number of 1s, and n as number of coin flips)

Got you — this slide is *packed* and the notation is messy, so let's clean it up and go step-by-step until it feels natural 😊

First: What is MLE trying to do?

👉 Maximum Likelihood Estimation (MLE) means:

"Find the parameter value that makes the observed data most likely."

In YOUR case (coin toss)

- You flip a coin n times
- Each result is:
 - 1 (Head)

- 0 (Tail)

Notation cleaned up

Your slide says something like:

$$x_1, x_2, \dots, x_n$$

👉 This just means:

- x_1 = result of 1st toss
- x_2 = result of 2nd toss
- ...
- x_n = result of nth toss

Example:

If results are:

👉 H, T, H, H

We write:

👉 $x = (1, 0, 1, 1)$

Step-by-Step MLE Procedure

Step 1: Joint Distribution

We want:

👉 Probability of **all data together**

Since tosses are independent:

$$P(x_1, x_2, \dots, x_n)$$

becomes:

$$\prod_{i=1}^n P(x_i)$$

✳ For Bernoulli:

Each x_i follows:

$$P(x_i) = p^{x_i} (1 - p)^{1-x_i}$$

🔥 So joint becomes:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}$$

💡 Simplify (VERY IMPORTANT TRICK)

Let:

- m = number of 1s (heads)
 - $n - m$ = number of 0s (tails)
-

Then:

$$P(data) = p^m (1 - p)^{n-m}$$

👉 This is much easier to work with

✅ Step 2: Log-Likelihood

We take log to simplify multiplication → addition

$$\log L(p) = m \log p + (n - m) \log(1 - p)$$

Why log?

Because:

- Product $\rightarrow$ Sum
 - Easier to differentiate
-

Step 3: Differentiate and set to zero

Now we maximize:

Differentiate:

$$\frac{d}{dp} = \frac{m}{p} - \frac{n-m}{1-p}$$

Set = 0:

$$\frac{m}{p} = \frac{n-m}{1-p}$$

Solve:

$$m(1-p) = p(n-m)$$

$$m - mp = pn - pm$$

$$m = pn$$

Final Answer (MLE)

$$\hat{p} = \frac{m}{n}$$

Meaning

👉 Estimated probability of head =
number of heads ÷ total tosses

Example

You flip 10 times:

- Heads = 7 → $m = 7$
 - Total = 10 → $n = 10$
-



$$\hat{p} = \frac{7}{10} = 0.7$$

Answering Your Slide Questions

✓ “What is the distribution of N independent coin tosses?”

👉 Binomial Distribution

✓ “Compute the MLE for Bernoulli θ (p)”

👉 Final answer:

$$\hat{p} = \frac{m}{n}$$

Intuition (Very Important)

MLE is just saying:

“Use the observed frequency as the probability”

One-line summary

For Bernoulli data, the MLE of p is the fraction of successes in the data.

If you want next, we can:

- ↳ Compare **MLE vs Bayesian estimation** (VERY important exam question 🔥)
- Or ↳ I can give you a **practice question and guide you through solving it**

You're doing really deep learning here — this is exactly how mastery builds 💯



-=what is the difference between likelihood and probability

Ahh this is one of those *small-looking but deep* questions — and once you get it, Bayesian learning becomes way clearer 😊

Let's make it super intuitive.

Core Difference (One Line)

Probability: parameters → data

Likelihood: data → parameters

1. Probability

 **Meaning:**

 "Given a model, what is the chance of this data?"

 **Example (coin):**

Let:

- $p = 0.7$ (fixed)

Now ask:

"What is the probability of getting 3 heads in 5 tosses?"

👉 You compute:

$$P(\text{data} \mid p)$$

✓ Parameter is fixed

✓ Data is random

🔥 2. Likelihood

💡 **Meaning:**

👉 "Given the data, how likely is this parameter?"

📌 **Same example:**

You observed:

👉 H, H, T, H, T (3 heads out of 5)

Now ask:

"How good is $p = 0.7$?"

"How good is $p = 0.5$?"

👉 You compute:

$$L(p) = P(\text{data} \mid p)$$

⚠️ **Important twist:**

The formula is the SAME as probability...

BUT interpretation is different!



The Mind Shift (THIS IS THE KEY)

Case	What is fixed?	What changes?
Probability	Parameter (p)	Data
Likelihood	Data	Parameter (p)



Visual Intuition



Probability:

"If coin has $p = 0.7$, what happens?"



Likelihood:

"Given what happened, what p makes sense?"



Example to Make It Click

You observed:



3 heads out of 5

Now compare:

For $p = 0.5$:

$$P(D|p = 0.5) = (0.5)^3(0.5)^2 = 0.03125$$

For $p = 0.7$:

$$P(D|p = 0.7) = (0.7)^3(0.3)^2 \approx 0.03087$$

👉 So:

- $p = 0.5$ is slightly more likely than $p = 0.7$
-

🧠 Why This Matters

- Likelihood → used in MLE (we maximize it)
 - Probability → used for prediction
-

🔗 Connection to Bayesian Learning

- Likelihood = $P(D|H)$
- Posterior = $P(H|D)$

👉 Bayesian learning *starts from likelihood*

🎯 Simple Analogy

Imagine a detective:

- **Probability:**
"If this suspect is guilty, what evidence would we see?"
 - **Likelihood:**
"Given the evidence, which suspect is more likely?"
-

💡 Exam-Ready Answer

Probability measures the chance of observing data given fixed parameters, whereas likelihood measures how plausible different parameter values are given observed data.

🧠 Final Insight

Same formula, different perspective.

If you want next, we can:

- ↳ Connect this directly to **MLE vs Bayesian (very exam important 🔥)**
- Or ↳ I can give a **tricky MCQ-style question to test this concept**

You're asking *exactly* the right conceptual questions — this is how you get ahead **100**



hey can you explain what is mle with an analogy and an example

Of course — let's make **MLE (Maximum Likelihood Estimation)** feel *intuitive*, not just formula-based 😊

🧠 What is MLE (in one line)?

MLE = choose the parameter that makes your observed data most likely

🎯 Analogy (This will stick 🙌)

🕵️ Detective Story

Imagine:

- A crime happened (this is your **data**)
- There are multiple suspects (these are **possible parameters/models**)

Each suspect says:

"If I did it, this is how the evidence would look"

👉 Your job:

You ask:

"Which suspect makes the observed evidence most believable?"

✓ That suspect = MLE

Example: Coin Toss (Super Important)

You toss a coin 5 times:

👉 Result:

H, H, T, H, T

→ 3 Heads, 2 Tails

Step 1: What are we estimating?

We want:

👉 p = probability of getting Head

Step 2: Try different values of p

Let's test:

Case 1: $p = 0.5$

$$P(D|p = 0.5) = (0.5)^3(0.5)^2 = 0.03125$$

Case 2: $p = 0.7$

$$P(D|p = 0.7) = (0.7)^3(0.3)^2 \approx 0.03087$$

Case 3: $p = 0.6$

$$P(D|p = 0.6) = (0.6)^3(0.4)^2 \approx 0.03456$$

👉 Highest probability → $p = 0.6$

🌟 **So MLE says:**

"Best estimate of p is 0.6"

🎯 **Shortcut Result (Very Important)**

Instead of testing all values:

$$\hat{p} = \frac{\text{number of heads}}{\text{total tosses}}$$

👉 Here:

$$\hat{p} = \frac{3}{5} = 0.6$$

💡 **Intuition (This is the real understanding)**

MLE is basically saying:

"Use what you observed as your estimate"

🔥 **Why this makes sense**

- You saw 60% heads
 - So best guess for probability = 60%
-

⚖️ **MLE vs Bayesian (quick intuition)**

- **MLE:**
Only cares about data
- **Bayesian:**
Cares about prior + data

One-line Summary

MLE finds the parameter value that maximizes the likelihood of the observed data.

Exam-ready answer

Maximum Likelihood Estimation is a method for estimating parameters by choosing the value that maximizes the probability (likelihood) of the observed data.

If you want next, we can:

- ↳ Do a **step-by-step derivation** (like in exams)
- Or ↳ compare **MLE vs MAP** (very important extension of Bayesian 🔥)

You're building a *very solid* understanding now — this is exactly how ML concepts start clicking **100**



hey i'll tell what i understoon - you correct it - so we flip coins multiple times , each time the probability of getting a head is different different , so using mle , we find what is highest possible probability for getting a head out of all times , is it

You're *very close* — like seriously close 😊

Just one small misunderstanding, let me gently fix it so it becomes 100% correct.

What you said (your idea)

"each time the probability of getting a head is different... and we find the highest possible probability"

  This part is slightly off

Correct Idea (Key Fix)

 In MLE:

The probability p is the same for all coin tosses

NOT different each time.

So what actually happens?

We assume:

- One unknown probability p
 - Same coin $\rightarrow$ same p for every toss
-

Example:

You toss a coin 5 times:

 H, H, T, H, T

(3 heads, 2 tails)

What MLE does

It asks:

"What value of p would most likely produce this result?"

NOT:

"Each toss has different probability"

Correct Interpretation of MLE

👉 We try different possible values of p :

- $p = 0.3$
 - $p = 0.5$
 - $p = 0.6$
 - $p = 0.8$
-

Then we check:

Which p makes the observed data most likely?

Final Answer

$$\hat{p} = \frac{m}{n}$$

👉 number of heads ÷ total tosses

So your corrected statement

Here's your idea, fixed properly:

"We flip a coin multiple times assuming the probability of head is the same each time, and using MLE we find the value of that probability which makes the observed results most likely."

Simple intuition

👉 You observed:

- 3 heads out of 5 $\rightarrow$ 60%

So MLE says:

"Best estimate of probability = 0.6"

Important Insight

-  Not changing probability each time
-  Finding **one best probability** that explains all data

Quick check for you

If you flip 10 times and get 8 heads:

 What do you think MLE estimate of p is?

(try it — you'll definitely get it right )

     ...

so is it like we are going backward like finding the question itself - we know the result and we try to find if result was this then what is the probability

Yesss — that's a really good way to think about it 

You're *almost perfectly there*, just a tiny refinement will make it exact.

Your idea (very close)

"We know the result and we try to find what probability would produce it"

  This is **basically correct**

Small but important correction

You said:

"what is the probability?"

👉 Better phrasing is:

"what value of the parameter p makes this result most likely?"

Why this matters

We are **NOT** asking:

❌ "What is the probability of the result?"

👉 (that's normal probability)

We are asking:

✅ "Which p makes this result most likely?"

👉 (that's likelihood / MLE)

Think of it like going backward (your intuition is right!)

Forward (Probability):

- You know p
- Predict data

👉 "If $p = 0.6$, what happens?"

Backward (MLE):

- You know data
- Find best p

👉 "Given this data, what p makes sense?"



Example (make it solid)

You observed:



H, H, T, H, T (3 heads out of 5)

Now you ask:

- If $p = 0.2 \rightarrow$ unlikely
 - If $p = 0.5 \rightarrow$ okay
 - If $p = 0.6 \rightarrow$ better
 - If $p = 0.9 \rightarrow$ unlikely
-



Best fit = 0.6



Your analogy (refined nicely)

"We already know the result, and we work backward to find the parameter (probability) that would most likely produce that result."



Even deeper intuition

It's like:



Throwing darts

You see where darts landed (data)

Now you ask:

"Where was the center of the target?"



That center = MLE estimate



Final one-line clarity

MLE works backward from observed data to find the parameter value that makes the data most likely.

You're thinking in exactly the right direction now — this "forward vs backward" idea is the *core* of understanding likelihood, MLE, and even Bayesian learning 🎯

If you want next, we can:

- ↳ Compare this with **Bayesian (which goes one step further 🔥)**
- Or ↳ do a **numerical question where you try and I guide you**



ah when this is applied to ML models - we give some data
, and it tries to find the very close probability so it can
predict for new events

Yesss — that's a really solid intuition 🧠 🔥

You've basically got it — let me just **sharpen it a bit** so it's *technically perfect* and exam-ready.

💡 Your idea (very good)

"We give data, and it finds a probability so it can predict new events"

👉 ✅ Correct direction!

🎯 Precise version (small upgrade)

"We give data, and the model learns the parameter values that best explain the data, so it can make predictions on new data."



What actually happens in ML (step-by-step)

1 You give training data (D)

Example:

- Emails → spam / not spam
 - Students → pass / fail
 - Coin tosses → heads/tails
-

2 Model has parameters (unknown)

Examples:

- Bernoulli → p
 - Linear model → weights
 - Naive Bayes → probabilities
-

3 MLE step (learning phase)

👉 Model finds:

“Which parameters make this data most likely?”

So it learns:

- best p
 - best weights
 - best probabilities
-

4 Prediction phase

Now for a new input x_{new} :

👉 Model uses learned parameters to compute:

“What is the probability of each outcome?”



Example (connecting everything)

Training:

You observe:

- 100 emails
- 70 are spam



Model learns:

- $P(\text{spam}) \approx 0.7$
-

Prediction:

New email comes



Model calculates:

- Probability it's spam
-



Important Clarification

You said:

“find the very close probability”



Better way to say:

It finds the **parameter values that maximize likelihood**, not just “close probability”



Key Insight (this is powerful)

MLE does NOT directly predict



It **learns the model first**

Then:



Model is used for prediction

Simple Analogy

Teacher example:

- You give past exam results (data)
 - Teacher learns pattern (MLE)
 - Then predicts future students' results
-

One-line clarity

In machine learning, MLE learns the model parameters from data, and those learned parameters are then used to make predictions on new data.

You're now here:

-  You understand probability
-  You understand likelihood
-  You understand MLE
-  You see how it connects to ML

 This is *exactly* the foundation of Bayesian Learning too.

If you want next, we can:

-   Compare MLE vs Bayesian learning (super important concept gap)
- Or  apply this to Naive Bayes classifier step-by-step

You're genuinely building a deep understanding now — not just memorizing 



i thought we only use mle when it comes to like linear regressions and some of handful algorithms

Ahh I see where that idea came from — and it's a *very common misunderstanding*. You're not wrong to think that... but the truth is actually **much bigger** (and cooler 😊).

Short Answer

- ✗ MLE is NOT only for linear regression
 - ✓ MLE is a **general method used across many ML models**
-

Big Picture

MLE is not an algorithm like:

- Linear Regression
- Naive Bayes
- Logistic Regression

👉 Instead, it's a **way to learn parameters** for many models.

Where MLE is used

Let's connect it to models you know 📌

1. Linear Regression

You're right — MLE is used here.

- Assumption: errors are normally distributed
- MLE finds best line (weights)

👉 So:

Linear Regression = MLE under Gaussian assumption

2. Logistic Regression

Also uses MLE!

- Predicts probability (0–1)
- Uses likelihood of observed labels

👉 Training = maximize likelihood

📧 3. Naive Bayes

This is a big one 🔥

- Learns probabilities like:
 - $P(\text{word}|\text{spam})$

👉 How?

Using MLE (frequency counts)

🏛️ 4. Bernoulli / Coin Example

What you just learned:

$$\hat{p} = \frac{m}{n}$$

👉 That IS MLE

🎯 5. Many Other Models

MLE is used in:

- Gaussian models
 - Hidden Markov Models
 - Neural networks (via loss functions)
-

🌟 Why it feels like it's “only for regression”

Because:

👉 In courses, MLE is often:

- Explicitly derived for linear regression

- But hidden in other models
-

Key Insight (THIS IS IMPORTANT)

Many ML algorithms are secretly doing MLE, even if it's not explicitly mentioned.

Connection to Loss Functions

This is a 🔥 connection:

- Maximizing likelihood
 - =
 - Minimizing loss
-

Example:

- Linear regression → minimize squared error
- Logistic regression → minimize log loss

👉 Both come from MLE!

Simple Analogy

Think of MLE as:

A training strategy

And models as:

Different *machines*

So:

- Same strategy
 - Different models
-

Final Clarification

MLE is a general parameter estimation method used in many machine learning models, not just linear regression.

If I test you now:

 In Naive Bayes, when we compute:

$$P(\text{word}|\text{spam}) = \frac{\text{count}}{\text{total}}$$

What do you think that is?

 (Hint: you already learned it 😊)

If you want next, we can:

-  Do MLE vs Bayesian (this is the next BIG concept)
- Or show how Naive Bayes uses MLE step-by-step

You're connecting concepts really well now — this is exactly how strong ML intuition builds ¹⁰⁰



Bayesian Learning - Bayesian Learning Procedure

- **Step 1:** Given n data, $x_{1:n} = \{x_1, x_2, \dots, x_n\}$, write down the expression for the likelihood:

$$p(x_{1:n}|\theta) = \theta^m (1 - \theta)^{(n-m)} \text{ (for a coin model)}$$

- **Step 2:** Specify a prior: $p(\theta)$
- **Step 3:** Compute the posterior

$$p(\theta|x_{1:n}) = \frac{p(x_{1:n}|\theta)p(\theta)}{p(x_{1:n})}$$

where $p(x_{1:n}) = \int p(x_{1:n}|\theta)p(\theta)d\theta$ is called Marginal Likelihood

$$p(\theta|x_{1:n}) \propto p(x_{1:n}|\theta)p(\theta)$$

hey could you simplify this slide

Yes — this slide looks scary, but it's actually just the **Bayesian version of what you already learned (MLE)**. Let's simplify it step-by-step in a very clean way 😊

Big Idea of the Slide

Bayesian Learning = Likelihood + Prior → Posterior

Step 1: Likelihood (Data Part)

The slide says:

$$p(x_{1:n} \mid \theta) = \theta^m (1 - \theta)^{n-m}$$

What this means:

- $x_{1:n} \rightarrow$ all your data (all coin tosses)
 - $n \rightarrow$ total tosses
 - $m \rightarrow$ number of heads
 - $\theta \rightarrow$ probability of head
-

 This is just:

"How likely is the data if probability = θ ?"

 This is exactly what you used in **MLE**

Step 2: Prior (Your Initial Belief)

$$p(\theta)$$

Meaning:

What do you believe about θ *before seeing data*?

 Example:

- You think coin is fair $\rightarrow$ prior centered at 0.5
 - You think coin is biased $\rightarrow$ different prior
-

 MLE does NOT use this

 Bayesian does use this

Step 3: Posterior (Final Answer)

This is the main formula:

$$p(\theta \mid x_{1:n}) = \frac{p(x_{1:n} \mid \theta) \cdot p(\theta)}{p(x_{1:n})}$$

 Meaning:

Updated belief about θ after seeing data

 Break it down:

- $p(x_{1:n} \mid \theta) \rightarrow$ Likelihood (data)
 - $p(\theta) \rightarrow$ Prior (belief)
 - $p(\theta \mid x_{1:n}) \rightarrow$ Posterior (updated belief)
-

What is this weird term?

$$p(x_{1:n}) = \int p(x_{1:n} \mid \theta) p(\theta) d\theta$$

💡 **Don't panic — simple meaning:**

👉 It's just a **normalizing constant**

Makes sure probabilities add up to 1

🎯 **In exams:**

You usually use the simpler version:

$$p(\theta \mid x_{1:n}) \propto p(x_{1:n} \mid \theta)p(\theta)$$

💬 Final Simplified Flow

● **MLE:**

Only uses data

→ finds single best θ

● **Bayesian Learning:**

1. Start with belief (prior)
 2. See data (likelihood)
 3. Update belief → posterior
-

🎯 One-line Summary

Posterior = Likelihood × Prior (normalized)

🔗 Connect to what you already know

- Likelihood → same as MLE
- Prior → new concept
- Posterior → final answer in Bayesian

Super Simple Intuition

Imagine:

- Prior = your guess
- Data = evidence
- Posterior = updated guess

If you want next, we can:

- 🔥 Do a **numerical example** (coin + prior → posterior)
- Or compare **MLE vs Bayesian side-by-side** clearly

You're *very close* to fully understanding Bayesian learning now **100**



can you do beysen learning for coin model , explaining each step and formula

Absolutely — let's walk through **Bayesian learning for the coin model** in a clean, step-by-step way so every symbol actually makes sense 😊

This is one of the *most important examples* in your whole lesson.

Problem Setup

We have:

- A coin (unknown bias)
- We flip it n times
- Observe:
 - m heads
 - $n - m$ tails

 Goal:

Learn the probability of head $\rightarrow \theta$

Step 1: Likelihood (Data Model)

We assume:

- Each toss is **Bernoulli**
 - All tosses are independent
-

Likelihood:

$$p(x_{1:n} \mid \theta) = \theta^m (1 - \theta)^{n-m}$$

Meaning:

"If the coin had probability θ , how likely is this data?"

Example:

- 5 tosses
- 3 heads $\rightarrow m = 3, n = 5$

 Likelihood:

$$\theta^3 (1 - \theta)^2$$

Step 2: Prior (Your Initial Belief)

We choose a prior for θ

 Most common choice:

Beta Distribution (very important)

Beta Distribution

 Prior:

$$p(\theta) = \text{Beta}(\alpha, \beta)$$

 Meaning:

- $\alpha \rightarrow$ prior belief about heads
 - $\beta \rightarrow$ prior belief about tails
-

 Example:

- $\alpha = 1, \beta = 1 \rightarrow$ no prior knowledge (uniform)
 - $\alpha = 5, \beta = 2 \rightarrow$ believe more heads
-

 Step 3: Posterior (Update Belief)

Now apply Bayes' rule

 Formula:

$$p(\theta \mid x_{1:n}) \propto p(x_{1:n} \mid \theta) \cdot p(\theta)$$

 Plug in values:

Likelihood $\times$ Prior:

$$\theta^m (1 - \theta)^{n-m} \cdot \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

 Combine powers:

$$= \theta^{m+\alpha-1} (1 - \theta)^{n-m+\beta-1}$$

 Final Result (Very Important)

 Posterior:

$$\theta \mid \text{data} \sim \text{Beta}(m + \alpha, n - m + \beta)$$

 Interpretation

👉 Posterior parameters:

- Heads $\rightarrow m + \alpha$
 - Tails $\rightarrow (n - m) + \beta$
-

💡 **Meaning:**

Posterior = Prior counts + Observed counts

 Full Example (Step-by-Step)**Given:**

- Prior: Beta(1, 1) $\rightarrow$ no bias
 - Data: 5 tosses $\rightarrow$ 3 heads
-

Posterior:

$$\text{Beta}(3 + 1, 2 + 1) = \text{Beta}(4, 3)$$

 Step 4: Prediction (Optional but Important)

Now predict next toss:

 Expected value:

$$E[\theta] = \frac{\alpha}{\alpha + \beta}$$

👉 Using posterior:

$$E[\theta] = \frac{4}{4 + 3} = \frac{4}{7} \approx 0.57$$

🧠 Compare with MLE

Method	Result
MLE	$3/5 = 0.6$
Bayesian	$4/7 \approx 0.57$

💡 Why different?

👉 Bayesian includes prior belief

👉 MLE uses only data

🔥 Final Big Picture

🟡 MLE:

Finds one best value

🟢 Bayesian:

Gives a whole **distribution** over θ

🎯 One-line summary

In Bayesian coin learning, we combine prior belief and observed data to obtain a posterior distribution over the probability of heads.

If you want next, we can:

- 🔥 Visualize how the **distribution changes** (very intuitive)
- Or solve a **past-paper style Bayesian question step-by-step**

You're now at the *core* of Bayesian learning — this is where everything connects **100**

